

InfraRisk: An open-source simulation platform for resilience analysis in interconnected power–water–transport networks

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ABSTRACT

Integrated simulation models are emerging as an alternative for analyzing large-scale interdependent infrastructure networks due to their modeling advantages over traditional interdependency models. This paper presents an open-source integrated simulation package for the component-level analysis of interdependent power-, water-, transport networks. The simulation platform, named 'InfraRisk' and developed in Python, can simulate network-wide effects of disaster-induced infrastructure failures and subsequent post-disaster restoration. InfraRisk consists of an infrastructure module, a hazard module, a recovery module, a simulation module, and a resilience quantification module. The infrastructure module integrates existing infrastructure network packages (*wntr* for water distribution systems, *pandapower* for power systems, and a static traffic assignment model for road transport systems) through an interface that facilitates the network-level simulation of infrastructure failures. The hazard module generates infrastructure component failures based on various disaster characteristics. The recovery module determines repair sequences and assigns repair crews based on predefined heuristics-based recovery strategies or model predictive control (MPC) based optimization. Based on the schedule, the simulation module simulates the consequences of the disaster impacts and the recovery actions on the performance of the interdependent network. The resilience quantification module offers system-level and consumer-level metrics to quantify both the risks and resilience of the integrated infrastructure networks against disaster events. InfraRisk provides a virtual platform for decision-makers to experiment and develop region-specific pre-disaster and post-disaster policies to enhance the overall resilience of interdependent urban infrastructure networks.

1. Introduction

Climate-induced extreme weather events and emerging threats, such as cyber-attacks and acts of terrorism, pose severe risks to the uninterrupted functioning of urban infrastructure systems.¹ While infrastructure systems, such as power-, water-, and transport systems, are critical for other infrastructure systems, businesses, and communities alike, they are also highly vulnerable to major hazards, such as earthquakes and floods. Of significant concern are the short- and long-term societal and economic losses resulting from infrastructure service disruptions, which may be compounded by the interdependencies among critical infrastructure systems. This had been evident in the recent 2021 Texas snowstorm, which caused power blackouts and water supply disruptions in the industrial hubs of Central Texas (Busby et al., 2021), leading to ripple effects in the national and global supply chains. Similar events in the past have shown that the scale of the indirect

economic impacts resulting from infrastructure disruptions are of magnitude larger than the direct costs incurred due to infrastructure failure and restoration (Hallegatte, Rentschler, & Rozenberg, 2019; National Research Council, 1999).

Empirical data and simulation studies confirm that judicious interventions to improve resilience against infrastructure failures resulting from external factors could not only reduce operational disruptions to individual infrastructure systems but also decrease the likelihood of large-scale network-wide consequences (de Almeida & Mostafavi, 2016; Sadashiva et al., 2021). Such interventions may be classified according to the targeted resilience characteristic(s) of the infrastructure system, namely resistant capacity, absorptive capacity, and restorative capacity (Ouyang, 2014; Vugrin, Warren, & Ehlen, 2011).

However, the challenge is to identify and evaluate various feasible resilience alternatives (both system-level and component-level) and

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¹ For ensuring consistency and clarity, we use the following conventions throughout the article: a 'system' refers to an infrastructure providing a certain utility service (for eg., a water or power distribution system). A 'network' is defined as a system of interconnected infrastructure systems (for eg., an interdependent power-, water-, and transport network.).

implement the cost-effective ones (Najarian & Lim, 2020). Moreover, such an effort requires frameworks and tools for a holistic understanding of the network-wide consequences of various system-level and component-level resilience strategies (Ouyang, 2014; Setola, Luijff, & Theocharidou, 2016) so that the cost and benefit aspects can be evaluated. In addition, the increasing societal and economic significance in infrastructure resilience studies also strengthens the case for developing simulation tools with the capability to perform more granular analyses.

While modeling and simulation tools have been extensively applied for resilience analysis in individual infrastructure systems, those intended for integrated infrastructure systems for component-level analyses are very few and require further research (Saidi, Kattan, Jayasinghe, Hettiaratchi, & Taron, 2018). Among the available integrated infrastructure models, there is little attention to interdependent water–power–transport networks, despite their considerable importance to the efficiency and resilience of cities. The overall objective of this study is to present an open-source simulation package for the analysis of interdependent power–water–transport network. The simulation platform is named ‘InfraRisk’ and is developed in Python. The specific objectives of InfraRisk are as follows:

1. Integrate existing domain-specific infrastructure simulation libraries using an object-oriented interface and present a sequential approach for the simulation of network-wide effects in interdependent power, water-, and transport systems.
2. Incorporate hazard initiation and recovery submodels in the integrated infrastructure model to investigate disaster-induced infrastructure component failures and subsequent post-disaster restoration actions on the resilience capabilities of the infrastructure systems.
3. Provide appropriate functional metrics to evaluate the effectiveness of pre- and post-disaster resilience interventions.

In summary, InfraRisk provides a decision-making support tool to design, test, and evaluate various component- and system-level resilience interventions in interdependent power-, water-, and transport systems using simulation approach. The rest of the paper is organized as follows: Section 2 provides a brief overview of the existing tools for interdependent infrastructure simulation; Section 3 discusses the architecture of InfraRisk simulation platform and its various functionalities; Section 4 presents an experimental simulation of interdependent effects of a few disaster scenarios on a test network using the InfraRisk platform, and Section 5 summarizes the conclusions.

2. Related work

While disaster resilience has been historically acknowledged as an indispensable quality of infrastructure systems, the need for considering the (inter)dependencies in assessing their ability to withstand unanticipated events has gained attention only in the last three decades. Until then, the focus was on developing best practices to enhance the physical security and robustness of individual infrastructure systems (Fisher, Norman, & Peerenboom, 2018). However, several disasters in the past, such as the World Trade Center attack in 2001, the Northeast blackout of 2003, the Indian Ocean earthquake and tsunami in 2004, and Hurricane Katrina in 2005, showed that the interconnections among critical infrastructure sectors intensified the societal and economic loss incurred by disasters (O’Rourke, 2007). Over the years, several countries incorporated infrastructure systems and their interdependencies as major considerations in aspects concerning national security and extended support for research and development in the area of infrastructure resilience (Communities Commission of the European, 2006; The White House, 2013). Consequently, academic interest in the identification, classification, and modeling of infrastructure interdependencies has grown considerably in the recent past, leading to the development of numerous frameworks, methods, and models for the analysis of interdependent infrastructure networks.

2.1. Traditional infrastructure simulation models

Traditional infrastructure interdependency models can be categorized into two groups: empirical models and computational models (Mitsova, 2021). Empirical models rely on data related to historical infrastructure breakdowns for characterizing the interactions between infrastructure systems. Empirical models could be built using qualitative data (for example, based on expert judgments Cooke & Goossens, 2004; Zorn, Pant, Thacker, Andreae, & Shamseldin, 2021) or quantitative data (for example, based on post-disaster impact assessment reports, newspaper reports, and official incident reports Luijff, Nieuwenhuijs, Klaver, van Eeten, & Cruz, 2009; McDaniel, Chang, Peterson, Mikawoz, & Reed, 2007). The major limitation of empirical models is their incapability to identify possible failure and cascading event scenarios that may not yet have occurred.

On the contrary, the computational approach explicitly captures physical-, cyber-, geographic-, and logical relationships among infrastructure systems using mathematical and logical functions along with functional data and attempts to mimic real-world systems in a controlled environment. Therefore, such models are effective in understanding the role of infrastructure properties and resilience interventions in system response. The most common computational models are graph-based models, system-dynamics models, and agent-based models.

In graph-based models, abstractions of infrastructure systems are created by representing infrastructure components producing or consuming resources using nodes and their interrelationships using edges (Svendsen & Wolthuisen, 2007, 2008). Graph theory and network-flow optimization concepts have been extensively applied to study the impacts of infrastructure component failures and their consequences (Holden, Val, Burkhard, & Nodwell, 2013; Praks, Kopustinskas, & Masera, 2017). The graph-based models do not consider detailed functional aspects of system components.

System dynamics-based models use the concepts of feedback loops, stocks, and flows to create abstractions of interdependent infrastructure networks. System dynamics-based models simulate the dynamic and evolutionary effects in the infrastructure system by testing various policies and investment alternatives using stock and flow variables (Pasqualini & Witkowski, 2005; Powell, DeLand, & Samsa, 2008). The notable drawbacks of system dynamics models are their incapability to analyze infrastructure systems at component-level, the need for a large amount of data for calibration, and excessive dependence on expert judgments for establishing feedback loops in the model (Ouyang, 2014).

The agent-based modeling (ABM) is a bottom-up approach for modeling interdependent infrastructure networks at a more granular scale by considering them as system-of-systems (SoS) (Nilsson & Darley, 2006). ABMs simulate infrastructure components as agents and allow them to interact with each other in representative operating environments to obtain quantitative insights into the system behavior (Balakrishnan & Zhang, 2020; Tesfatsion, 2003; Thompson et al., 2019). Some of the key issues pertaining to ABM are its over-dependence on assumptions, calibration of parameters, and validation of the results (Ouyang, 2014).

In a few other models, a hybrid approach that combines empirical models with computational models have been proposed. For example, Cavalieri, Franchin, Gehl, and D’Ayala (2017) applied simulation techniques to generate conditional probability tables for constructing Bayesian Networks for assessing seismic risks in water networks. In addition to empirical and computational models, economic theory models, such as input–output models (Haimes et al., 2005; Oliva, Panzieri, & Setola, 2010), have also been extensively used for modeling interdependencies. However, such models, like empirical models, depend on disaster data for calibrating the strength of dependencies among infrastructure sectors.

Table 1
Examples of water, power and traffic simulation packages and tools.

Infrastructure systems	Simulation platform	Programming language	Open-source	Functionalities
Water	EPANET and its extensions	Standalone with GUI, versions in Python (wntr), R (Epanet2toolkit), Matlab (EPANET Toolkit).	Yes	Hydraulic simulation of water distribution systems.
	WaterGEMS	Standalone with GUI	No	Hydraulic simulation of water distribution systems
	Replica	Standalone with GUI	No	Dynamic simulation of fluids, including hydraulic simulation.
	TSNet	Python	Yes	Transient simulation of water distribution systems.
Power	Matpower	MATLAB	Yes	Steady-state power system simulation and optimization.
	Pandapower	Python	Yes	Analysis and optimization of power systems.
	PowerWorld Simulator	Standalone with GUI	No	Interactive simulation, analysis and visualization of power systems.
	OpenDSS	Standalone with GUI	Yes	Simulation of electric utility distribution systems.
	GridLAB-D	Standalone with GUI	Yes	Simulation and analysis of power distribution systems.
	Dome	Python	Yes	Power system analysis.
Transportation	MATSIM	Java	Yes	Agent-based simulation of large-scale transport systems, including road networks.
	MatLAB DTA package	MATLAB	Yes	Dynamic traffic assignment in road traffic networks.
	Vissim	Standalone with GUI	No	Microscopic and mesoscopic traffic simulation.
	Python STA package	Python	Yes	Static traffic assignment in road traffic networks.

2.2. Integrated infrastructure simulation models

As an alternative to traditional interdependency models, recent advancements in computational modeling have led to the integration of existing infrastructure-specific simulation models (Eusgeld, Nan, & Dietz, 2011; Nan & Eusgeld, 2011; Portante, Kavicky, Craig, Talaber, & Folga, 2017). Such integrated models intend to take full advantage of existing domain-specific models to simulate the global behavior of complex heterogeneous systems (Gomes, Thule, Broman, Larsen, & Vangheluwe, 2018). The key challenges of developing an integrated infrastructure model are time synchronization and coupling of constituent simulators having divergent functional dynamics, complexity, and time-scale (Caire, Sánchez, & Hadjsaid, 2013).

As far as critical infrastructure systems are concerned, integrated models have been used in various fields. For example, Monti, Colciago, Conti, Maglio, and Dougal (2009) developed an integrated model by combining VTBPro (power system), Matlab Simulink (control system) and OpNet (communication system) simulation tools for studying the role of communication systems in the performance of power grids. In another study, Erdener, Pambour, Lavin, and Dengiz (2014) developed an electricity and gas network simulation and analysis platform by integrating infrastructure specific simulation models using a MATLAB interface.

A more advanced set of integrated models, known as co-simulation models, adopt a distributed simulation approach. Co-simulation frameworks can be used to operate computationally intensive domain-specific submodels on different computers and seamlessly interact with each other on a real-time basis, enhancing both efficiency and usability (Wang, Magoua, & Li, 2022). Several frameworks and standards have been developed to implement co-simulation of complex systems, such as functional mock-up interface (FMI), high-level architecture (HLA), and distributed interactive simulation (DIS) (Ahmad, Kazmi, Shahzad, Palensky, & Gawlik, 2016). In recent years, the high-level architecture (HLA) standard has been increasingly adopted to develop interdependent infrastructure models for perform simulations in a distributed environment (Dubaniowski & Heinimann, 2019; Wang et al., 2022).

Recent studies have also extended the scope of integrated infrastructure models by combining infrastructure-specific simulators with other urban system models for investigating various aspects of disaster resilience of urban regions. For example, Marasco et al. (2021) developed a Python-based integrated platform for assessing the vulnerability and resilience of urban power and water systems against seismic events by combining infrastructure simulators and agent-based socio-technical models. Yang et al. (2021) combined geographic information system (GIS) tools, building information modeling (BIM) tools, and domain-specific infrastructure simulation models to develop an integrated model to analyze the risks of urban flooding on both infrastructure and communities. Similarly, Franchin and Cavalieri (2015) studied the resilience characteristics of urban infrastructure and communities against seismic hazards by integrating existing seismic models, a network-based urban infrastructure network model (utilities, critical facilities, and build), and a community vulnerability model.

2.3. Simulation of power-, water-, and transport systems and their interdependencies

As far as power-, water-, and transport systems are concerned, there are several standalone simulation tools that can replicate the real-world operational characteristics. Table 1 lists some of the most common software packages used for the simulation of power-, water-, and transport systems and their operations. In the case of water distribution systems, most of the available tools adopt hydraulic modeling (for e.g., EPANET), which utilize concepts from areas, such as network simulation, statistical analysis, and classical optimization methods (Berglund et al., 2020). In power system simulation, most of the tools focus on power flow optimization (steady state and dynamic), stability analysis, and short circuit detection (eg. *pandapower*), which are modeled and solved as a combination of algebraic and differential equations (Vugrin & Baca, 2013). Traffic flow on road networks are modeled and simulated by different methods, such as network-flow optimization techniques (static traffic assignment and dynamic traffic assignment),

multi-agent simulations models (eg. MATSIM), microscopic/mesoscopic simulation (eg. VISSIM), and cellular automata (Kotusevski & Hawick, 2009).

In addition, several studies have attempted to model the dependencies in interdependent power-water, power-transport and water-transport networks, using traditional- and integrated simulation approaches. The major component-level dependencies between water and power systems commonly modeled are pumping stations and storage tanks (electric power to operate water pumps) (Dudenhoeffer et al., 2007; Zhang, Yang, & Lall, 2016), and cooling stations in power plants (water required as coolant) (Thompson et al., 2019). The most commonly found dependency between transport system and other two systems is related to the road accessibility after a hazard occurs. Cascading failures resulting from fallen electric poles (Zou & Chen, 2019) and damaged pipelines/sewers (Dong, Wang, Mostafizi, & Song, 2020) road accessibility have been modeled and studied in the past.

2.4. Research gaps

The following are the major findings from the review of the literature related to simulation of interdependent infrastructure networks.

1. Most of the available traditional interdependent infrastructure simulation models provide a system-level abstraction of infrastructure system performance with limited emphasis on the component-level operational characteristics.
2. Due to their intrinsic limitations and homogeneous modeling approaches, traditional simulation models are incapable of leveraging upon the existing domain knowledge in the constituent infrastructure sectors. Consequently, their ability to perform analyses involving realistic resilience interventions is often limited.
3. Recent advancements in integrated modeling techniques have led to its greater adoption in interdependent infrastructure modeling and analysis. However, the use of integrated models in infrastructure analysis is an emerging field and in the early stages of adoption, especially in the case of civil infrastructure systems.

Several disasters, such as, Hurricane Harvey in 2017 and Texas snowstorm in 2021, have shown how critical power-, water-, and transport systems are for economic and community resilience in cities (Doss-Gollin, Farnham, Lall, & Modi, 2021; Frame, Wehner, Noy, & Rosier, 2020). The uninterrupted and coordinated functioning of the above systems is vital for both normal operations and post-disaster recovery. Despite the relevance of the above systems and their interdependencies in disaster resilience of cities, there is a lack of integrated infrastructure tools for analyzing interdependent power-, water-, and transport network. An open-source integrated platform to simulate disaster impacts on large-scale interdependent power-, water-, and transport networks could be of interest to a wide range of stakeholders involved in urban resilience decision-making and help in advance the use of integrated models for the analysis of other interdependent infrastructure networks. The integration of individual infrastructure simulators in InfraRisk is performed using a simpler sequential approach compared to the distributed simulation frameworks, such as HLA and DIS, while maintaining adequate simulation performance.

3. Methodology

3.1. Methodological framework

Fig. 1 illustrates the methodological framework adopted in the InfraRisk simulation platform. The platform is based on the widely accepted risk- and resilience analysis frameworks as presented in Argyroudis et al. (2020) and Balakrishnan (2020). In this framework, the most important component is an interdependent infrastructure model

that consists of various infrastructure system submodels of interest. In addition, the major hazards in the region can also be modeled. Further, the vulnerabilities in the network to those hazards are mapped and the direct impacts (physical and functional failures in the infrastructure components) are simulated using the hazard model. For scheduling post-disaster restoration/repair actions, a recovery model is also developed. The restoration actions are prioritized based on specific recovery strategies or optimization methods. The indirect failures in the network are simulated using the interdependent infrastructure model based on the initial failure events and the subsequent repair actions. The component- and system-level operational performance are quantified and tracked using appropriate resilience metrics.

3.2. Implementation in Python

Python programming language is chosen for developing InfraRisk because of its versatility, ease of use, and extensive open-source libraries. Python repositories house several efficient libraries to model individual infrastructure systems, such as water (*wntr*) and power (*pypower* and *pandapower*) systems, which could be adopted as domain-specific infrastructure simulators.

The basic idea behind the InfraRisk simulation package is to integrate existing infrastructure-specific simulation models through an object-oriented interface so that interdependent infrastructure simulation can be achieved. Interfacing requires identifying and modeling the dependencies among various infrastructure components and time-synchronization among infrastructure simulation models. To address the above challenges, InfraRisk is built using a sequential simulation framework (Fig. 1). The advantage of this approach is that it simplifies the efforts for data preparation and enables the complete utilization of component-level modeling features of the domain-specific infrastructure models.

InfraRisk consists of five modules, namely, (a) integrated infrastructure network simulation, (b) hazard initiation and vulnerability modeling, (c) recovery modeling, (d) simulation of direct and indirect effects, and (e) resilience quantification. In the rest of the section, a detailed discussion on each of the above modules is provided.

3.2.1. Integrated infrastructure network simulation

This module houses the three infrastructure models to simulate power-, water-, and transport systems. These models are developed using existing Python-based packages. In order to model the power system, *pandapower* is employed (Thurner et al., 2018). The water distribution system is modeled using *wntr* package (Klise et al., 2020). The traffic model provides the travel costs for traveling from one point in the network to another and is modeled using the static traffic assignment method (Boyles, Lowmes, & Unnikrishnan, 2020). All three packages have network-flow optimization models that identify the steady-state resource flows in the respective systems considering the operational constraints. The details of the packages are presented in Table 2.

The *pandapower* package can be used to determine the steady-state optimal power flow for a given set of system conditions. The optimal power flow problem, solved by *pandapower*, attempts to minimize the total power distribution costs in the system under load flow-, branch-, bus-, and operational power constraints (Eq. (1)) (Thurner et al., 2018).

$$\begin{aligned}
 \min \quad & \sum_{i \in \text{gen}, \text{sgen}, \text{load}, \text{extgrid}} P_i \times f_i(P_i) \\
 \text{s.t.} \quad & P_{\min,i} \leq P_i \leq P_{\max,i} \quad \forall i \in \text{gen}, \text{sgen}, \text{extgrid}, \text{load} \\
 & Q_{\min,i} \leq Q_i \leq Q_{\max,i} \quad \forall i \in \text{gen}, \text{sgen}, \text{extgrid}, \text{load} \\
 & V_{\min,j} \leq V_j \leq V_{\max,j} \quad \forall j \in \text{bus} \\
 & L_k \leq L_{\max,k} \quad \forall k \in \text{trafo}, \text{line}, \text{trafo3w}
 \end{aligned} \tag{1}$$

where i , j , and k are the power system components, gen is the set of generators, sgen is the set of static generators, load is the set of loads,

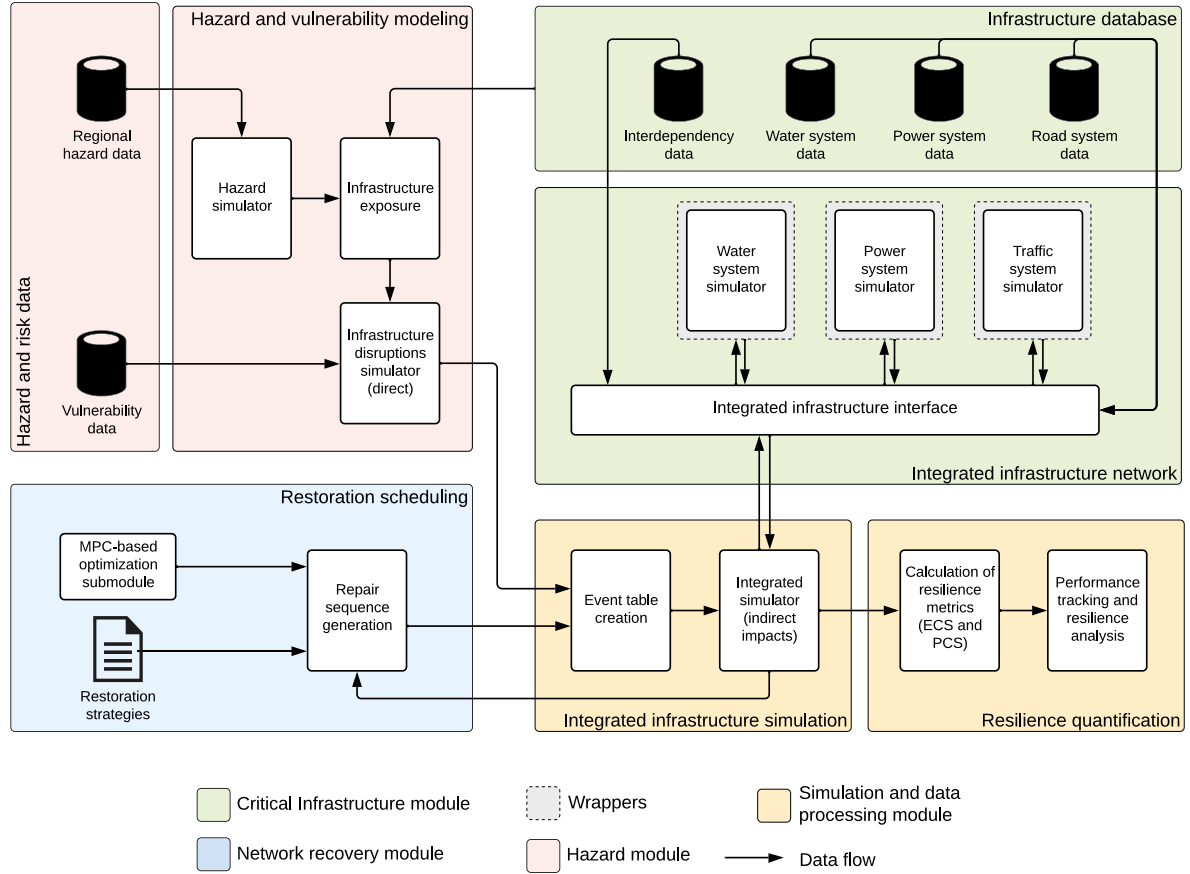


Fig. 1. Software implementation framework adopted in the InfraRisk simulation platform.

Table 2
Infrastructure packages using in the simulation model.

Infrastructure	Package	Capabilities
Power	<i>pandapower</i>	- Capable of generating power distribution systems with standard components, such as lines, buses, and transformers based on design data. - Capable of performing power-flow analysis.
Water	<i>wntr</i>	- Capable of generating water distribution systems with standard components such as pipes, tanks, and nodes based on design data. - Capable of performing pressure dependent demand or demand-driven hydraulic simulations.
Transport	Static traffic assignment package	Capable of implementing static traffic assignment and computing travel times between origin–destination pairs.

extgrid is the set of external grid connections, *bus* is the set of bus bars, *trafo* is the set of transformers, *line* is the set of lines, and *trafo3w* is the set of three winding transformers, $f_i(\cdot)$ is the cost function, P_i is the active power in i , Q_i is the reactive power in i , V_j is the voltage in j and L_k is the loading percentage in k .

The *wntr* package can simulate water flows in water distribution systems using two common approaches, namely, demand-driven analysis (DDA) and pressure-dependent demand analysis (PDA). While DDA assigns pipe flows based on the demands irrespective of the pressure at demand nodes, PDA assumes that the demand is a function of the pressure at which water is supplied. The PDA approach is more suitable

for pressure-deficient situations, such as disaster-induced disruptions to water infrastructure. In the case of PDA, the actual node demands is computed as a function of available the water pressure at the nodes as in Eq. (2) (Klise et al., 2020).

$$d_i(t) = \begin{cases} 0 & p_i(t) \leq P_0 \\ D_i(t) \left(\frac{p_i(t) - P_0}{P_f - P_0} \right)^{\frac{1}{\epsilon}} & P_0 < p_i(t) \leq P_f \\ D_i & p_i(t) > P_0 \end{cases} \quad (2)$$

where $d_i(t)$ is the actual demand at node i at time t , $D_i(t)$ is the desired demand at a node i at t , $p_i(t)$ is the available pressure in node i at t , P_f is the nominal pressure, and P_0 is the lower pressure threshold, below which no water is consumed. In InfraRisk, the hydraulic simulation is performed using the PDA approach.

The traffic condition in the transport system is modeled using the static traffic assignment method based on the principle of user-equilibrium. Under user-equilibrium, every user tries to minimize their travel costs. The traffic assignment problem considered in InfraRisk package is formulated as follows (Eq. (3)) (Boyles et al., 2020).

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{h}} \quad & \sum_{(i,j) \in A} \int_0^{x_{ij}} t_{ij}(x_{ij}) dx \\ \text{s.t.} \quad & x_{ij} = \sum_{\pi \in \Pi} h^\pi \delta_{ij}^\pi \quad \forall (i,j) \in A \\ & \sum_{\pi \in \Pi^{rs}} h^\pi = d^{rs} \quad \forall (r,s) \in Z^2 \\ & h^\pi \geq 0 \quad \forall \pi \in \Pi \end{aligned} \quad (3)$$

where A is the set of all road links with i and j as the tail and head nodes, t_{ij} is the travel cost on link (i,j) , x_{ij} is the traffic flow on link (i,j) , h^π is the flow on path $\pi \in \Pi$, δ_{ij}^π is an indicator variable

that denotes whether (i, j) is part of π , d^{rs} is the total flow between origin–destination pair r, s .

The module also consists of an interdependency layer which serves as an interface between infrastructure systems. The interdependency layer stipulates the different pieces of information that can be exchanged among individual infrastructure systems and their respective formats. The interdependency submodule also stores information related to the various component-to-component couplings between infrastructure systems. The module facilitates the communication between infrastructure systems and enables information transfer triggered by dependencies. Currently the following dependencies are considered.

- Power–water dependencies, which include dependency of water pumps on electric motors and generators on reservoirs (hydro-power).
- Dependencies also exist between road traffic system and the other two infrastructure systems, as the former provides access to the latter. The disruptions to transport infrastructure components and their recovery are key considerations that influence the restoration and recovery of all other infrastructure systems. The module also stores the functional details of all infrastructure components, including their operational status after a disaster.

The interdependency layer communicates with the infrastructure simulators through inbuilt functions (wrappers).

3.2.2. Hazard initiation and vulnerability modeling

The hazard module generates disaster scenarios and initiates disaster-induced infrastructure failures based on their vulnerability. The hazard initiation and the resulting infrastructure component failures is the first step in the interdependent infrastructure simulation. The probabilistic failure of an infrastructure component is modeled as follows (Eq. (4)):

$$p(\text{failure}_i) = p(\text{hazard}) \times p(\text{exposure}_i | \text{hazard}) \times p(\text{failure}_i | \text{exposure}_i) \quad (4)$$

where i is the component, $p(\cdot)$ is the probability. The probability of failure of a component is computed as the product of the probability of the hazard, the probability of the component being exposed to the hazard if it occurs, and the probability of failure of the component if it is exposed to the event.

The hazard module generates infrastructure component failures based on the hazard characteristics. In its current version, point events (such as random failures, terrorist attacks, and fire incidents), track-based events (such as floods, cyclones, and tornadoes), random events, and custom events can be generated using the module.

For generating point events, three parameters, namely, the location of occurrence of the event, the radius of the impact, and the intensity of the event are to be provided. The hazard module allows five predefined intensity levels — low, moderate, high, extreme, and random. The conditional failure probability ($p(\text{failure}_i | \text{exposure}_i)$) is dependent on the intensity level. The conditional probability of a component being exposed to the event ($p(\text{exposure}_i | \text{hazard})$) is defined as a function of the ratio of the distance of the component to the location of occurrence of the event and the radius of impact.

In the case of track-based events, the input parameters are the track of the event, intensity of the event, and the offset distance of impact. In the case of flood events, the GIS data of the regional water bodies are also needed. For other track-based events, such as tornadoes and hurricanes, the tracks are randomly generated using a spline function. For all track-based events, the conditional probability of being exposed to the event is calculated as a function of the ratio of the perpendicular distance from the component to the track to the offset distance of impact.

The third type of disruptions that can be generated is the random failure events. The user can choose the number of components that needs to be randomly failed in each disaster scenario.

In addition to the above three pre-defined disruption classes, InfraRisk includes a custom events class, which allows the user to define the disaster and failed components before performing the integrated network simulation. Due to its flexibility, this class of disruptions can be coupled with realistic hazard and vulnerability analysis models, such as Espinoza, Panteli, Mancarella, and Rudnick (2016), Pitilakis, Crowley, and Kaynia (2014) and Pregnotato, Ford, Wilkinson, and Dawson (2017).

A few disaster scenarios generated by the hazard module are illustrated in Fig. 2. The radius/offset distance of impact is marked in gray color, whereas the failed components as determined by Eq. (4) is denoted using red color.

After a disaster scenario is generated, the module stores the details of infrastructure disruptions (disrupted components, time of disruption, and the severity of disruption) in the local directory for use during the network simulation.

3.2.3. Recovery modeling

The third module, which is the recovery module, determines how the repair actions are sequenced and implemented. The three major factors that influence recovery are the availability of repair crews, repair times of components, and the criteria used for selecting subsequent components to restore. In InfraRisk, the user can specify the number of crews deployed for restoration of the three infrastructure systems, their initial locations in the network, and the repair times of the infrastructure components.

The repair sequence can be derived using two approaches. The first approach is to adopt pre-defined repair strategies based on performance- and network-based heuristics. Currently, there are three inbuilt strategies based on the following criteria:

1. Maximum flow handled: The resource-flow during normal operating conditions could reflect the importance of an infrastructure component to the system (Nicholson, Barker, & Ramirez-Marquez, 2016). The maximum resource-flow handled by a component, considering the temporal fluctuations, can be used as a performance-based heuristics to prioritize failed components for restoration.
2. Betweenness centrality: Centrality is a graph-based measure that is used to denote the relative importance of components (nodes and links) in a system. Betweenness centrality is often cited as an effective measure to identify critical infrastructure components (Almoghathawi & Barker, 2019).
3. Landuse/zone: Certain regions of a network may have consumers with large demands or critical to the functioning of the whole city. Industrial zones and central business districts are critical from both societal and economic perspectives.

While it is comparatively easier to derive repair sequences based on heuristics, they may not guarantee optimal recovery of the system or the network.

The second approach is an optimization model leveraging on the concept of model predictive control (MPC) (Camacho & Bordons, 2007). In this approach, first, out of n repair steps, the solution considering only k steps (called the prediction horizon) is computed. Next, the first step of the obtained solution is applied to the system and then the process is repeated for the remaining $n - 1$ components until all components are scheduled for repair. In the context of the integrated infrastructure simulation, the optimizer module evaluates repair sequences of the length of the prediction horizon for each infrastructure (assuming that each infrastructure has a separate recovery crew) based on a chosen resilience metric (Kottmann, Kyriakidis, Dang, & Sansavini, 2021) (see Section 3.2.5 for a detailed discussion on the

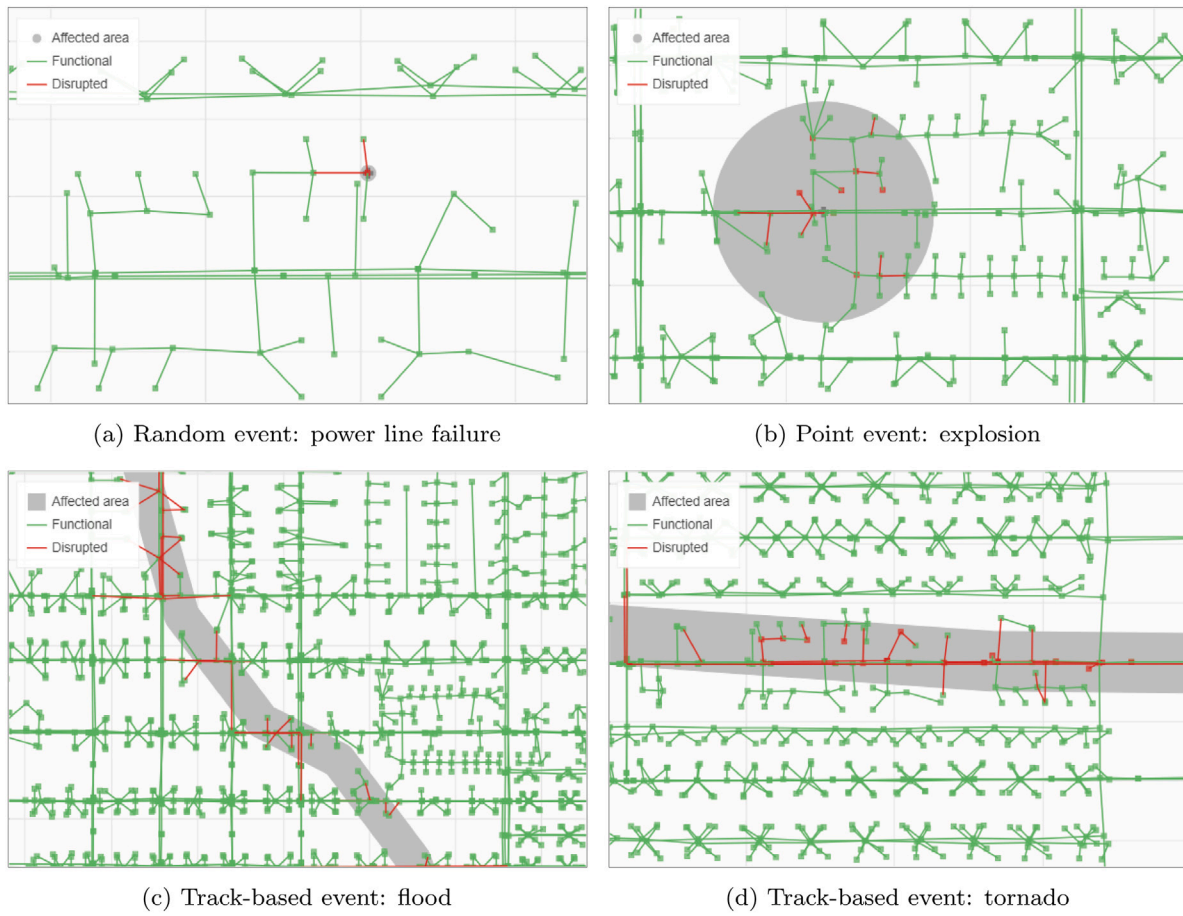


Fig. 2. Examples of disaster scenarios and resultant infrastructure failures generated by the hazard module.

resilience metrics used). The optimal repair sequence is found by maximizing the resilience metric. At this stage, the optimal repair action in each prediction horizon is computed using a brute-force approach where the resilience metric is evaluated for each of the possible repair sequences. The major limitation of MPC is that it is suitable only for small disruptions involving a few component failures; MPC becomes computationally expensive to derive optimal restoration sequences for larger disruptions due to the large number of repair permutations it has to simulate.

3.2.4. Simulation of direct and indirect effects

The simulation module implements the integrated infrastructure simulation in two steps, namely, event table generation and interdependent infrastructure simulation. The objective of the event table is to provide a reference object to schedule all the disruptions and repair actions for implementing the interdependent network simulation.

While individual infrastructure agencies may have finalized a restoration plan immediately after a disaster (especially one based on the strategies defined by heuristics), there are several factors that could influence its implementation. The most important hurdle is accessing the locations of disrupted components, as roads could also be disrupted either due to damage or debris (Comes & Van de Walle, 2014; Iloglu & Albert, 2020). In such a scenario, the crews may modify the repair sequence by prioritizing those components which are accessible under prevalent road system conditions. The modified repair sequence considering road access and the corresponding repair start and end times are computed by the model using the algorithm illustrated in Fig. 3.

The algorithm first checks if there are components affected by the disruptive event. If there are any, the infrastructure repair sequences

based on the strategy chosen by the user is derived and the repair tasks are assigned to the respective infrastructure system crews. Now, for each of the infrastructure system, one component at a time is selected in the order and its accessibility by the repair crew from their current location is ascertained. If the component is accessible, the shortest path (based on the static traffic assignment method under prevailing road conditions) is identified and the travel time is computed. Next, the start and the end times of the repair action are computed based on the travel time and the component repair time. At the end of the repair, if the component belongs to the transport system, the static traffic assignment model is rerun and the link flows and travel times are updated. This completes one cycle of the algorithm and the same process is repeated until all the components are scheduled for repair.

The component failures, repair actions, and the respective time-stamps, as obtained from the algorithm described in Fig. 3, are recorded in an event table for later use in the simulation module. The simulation platform uses the event table as a reference to modify the operational status of system components during the simulation, so that the consequences of disaster events and repair actions are reflected in the system performance. The recovery module also stores details including the number of repair crews for every infrastructure system, and their initial locations.

The next step is to simulate the interdependent effects resulting from the component disruptions and the subsequent restoration efforts. One of the main challenge in simulating the interdependent effects using a platform that integrates multiple infrastructure models is the time synchronization. Time synchronization is essential in InfraRisk for the following reasons:

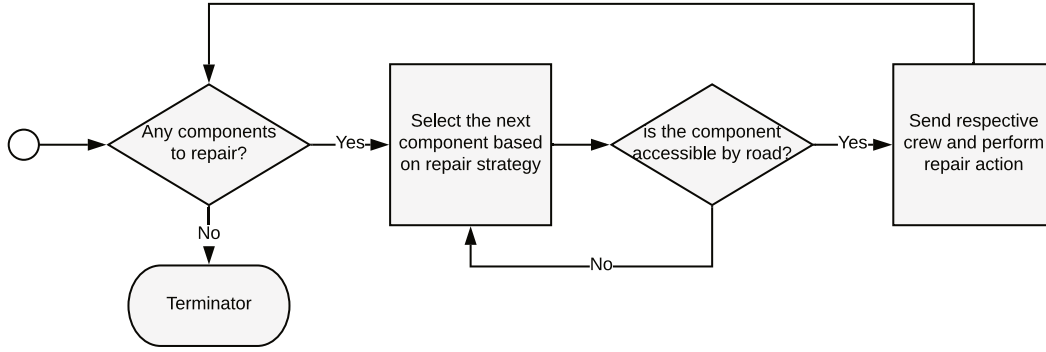


Fig. 3. Algorithm used for scheduling the repair actions scheduled using a predefined recovery strategy.

- The water system model (*wntr*) simulates the water flows during specified intervals, whereas the power system model *pandapower* computes power flows for a given time instance.
- The network-wide effects are immediately reflected in the power system, whereas, in the case of water distribution system, there might be delayed responses, such as gradual emptying of the tanks, which need to be captured by the resilience metrics.

In order to synchronize the times, the power- and water system models are run successively for every subsequent time-interval in the event table. The required water system metrics are collected for every one minute of simulation time from the *wntr* model, whereas power system characteristics at the start of every time interval is recorded from the *pandapower* model. The power flow characteristics are assumed to remain unchanged unless there is any modification to the power system in the subsequent time-steps in the simulation.

Algorithm 1 presents the pseudo-code for simulating the interdependent effects in the water and power systems.

Algorithm 1 Pseudo-code for simulating interdependent effects in water and power systems and calculating resilience metrics

```

timeList[] = sorted array of unique time stamps in the event table.
for i = 0 to timeList.length - 1 do
    Update operating status of the failed components at timeList[i] in both
    the power and water system models.
    Run the power system model and calculate the system performance at
    timeList[i].
    Update the indirectly affected components in the water system if any
    based on the predefined dependencies.
    Run the water system model for the time interval timeList[i] and
    timeList[i+1] and calculate the system performance.
    Update the indirectly affected components in the power system if any
    based on the predefined dependencies.
end for
Calculate the resilience metrics from the network performance time series.
  
```

3.2.5. Resilience quantification

The network-wide effects of infrastructure disruptions and the subsequent repair actions are measured using inbuilt resilience metrics. The repair actions are reflected in the resilience metric, which in turn can be used to identify the best resilience strategy or intervention.

Currently, the model has two measures of performance (MOP), namely, equitable consumer serviceability (ECS) and prioritized consumer serviceability (PCS), to quantify the system- and network steady-state performances. The above MOPs are based on the well-known concepts of satisfied demand (Didier, Broccardo, Esposito, & Stojadinovic, 2017) and productivity (Poulin & Kane, 2021). The MOPs are used as the basis for defining the resilience metrics.

Consider an interdependent infrastructure network $\mathbb{K}$ consisting of a set of infrastructure systems denoted by K : $K \in \mathbb{K}$. There are

N consumers who are connected to $\mathbb{K}$ and the resource supply from a system K to consumer $i \in N$ at time t under normal operating conditions is represented by $S_i^K(t)$.

The ECS approach assumes equal importance to all the consumers dependent on the system irrespective of the quantity of resources consumed from the system. For an infrastructure system, the ECS at time t is given by Eq. (5).

$$ECS_K(t) = \left(\sum_{\forall i: S_i^K(t) > 0} \frac{s_i^K(t)}{S_i^K(t)} \right) / n_K(t), \quad \text{where } 0 \leq s_i^K(t) \leq S_i^K(t) \quad (5)$$

where s_i is the resource supply at time t under stressed system conditions and $n_K(t)$ is the number of consumers with a non-zero normal demand at time t .

In the case of PCS, the consumers are weighted by the quantity of resources drawn by them. This approach assumes that disruptions to serviceability of large-scale consumers, such as manufacturing sector, have larger effect to the whole region compared to small-scale consumers such as residential buildings. The PCS metric of an infrastructure system at time t is given by Eq. (6).

$$PCS_K(t) = \left(\frac{\sum_{\forall i: S_i^K(t) > 0} s_i^K(t)}{\sum_{\forall i: S_i^K(t) > 0} S_i^K(t)} \right), \quad \text{where } 0 \leq s_i^K(t) \leq S_i^K(t) \quad (6)$$

The normal serviceability component ($S_i^K(t)$) makes both ECS and PCS metrics unaffected by the intrinsic design inefficiencies as well as the temporal fluctuations in demand.

For water distribution systems, pressure-driven approach is chosen as it is reported to be most ideal for the hydraulic simulation under pressure deficient situations. The component resource supply values for water systems are computed as in Eqs. (7)–(8).

$$s_i^{water}(t) = Q_i(t) \quad (7)$$

$$S_i^{water}(t) = Q_i^0(t) \quad (8)$$

where $Q_i(t)$ and $Q_i^0(t)$ are the water supplied to consumer i during stressed and normal system conditions, respectively.

For power systems, the power supplied to components under normal and stressed system conditions can be calculated using Eqs. (9)–(10).

$$s_i^{power}(t) = p_i(t) \quad (9)$$

$$S_i^{power}(t) = p_i^0(t) \quad (10)$$

where $p_i(t)$ and $p_i^0(t)$ are the power supplied to consumer i under stressed and normal power system conditions.

The ECS and PCS time series can be used to profile the effect of the disruption on any of the infrastructure systems. To quantify the system-level cumulative performance loss, a resilience metric called Equivalent Outage Hours (EOH), based on the well-known concept of 'resilience

Table 3
Component details of the simple network.

Power system		Water system		Traffic system	
Component	Count	Component	Count	Component	Count
Buses/poles	9	Main pipelines	12	Arterials	22
Loads	3	Demand nodes	9	Attractor zones	9
Motors	1	Pumps	1	Generator zones	9
Switches	0	Tanks	1		
External grids	1	Reservoirs	1		
Power lines	5				
Transformers	2				

triangle' (Bruneau et al., 2003), is introduced. EOH of an infrastructure system due to disaster event γ^K is calculated as in Eq. (11).

$$\gamma^K = \frac{1}{3600} \int_{t_0}^{t_{max}} [1 - PCS_K(t)] dt \quad \text{or} \quad \gamma^K = \frac{1}{3600} \int_{t_0}^{t_{max}} [1 - ECS_K(t)] dt \quad (11)$$

where t_0 is the time of the disaster event in the simulation and t_{max} is the maximum simulation time (both in seconds). In Eq. (11), system performance during normal operating conditions is 1 due to the expression of the MOP used (see Eqs. (5)–(6)).

EOH of an infrastructure system can be interpreted as the duration (in hours) of a full infrastructure service outage that would result in an equivalent quantity of reduced consumption of the same service by all consumers during a disaster. The larger the EOH value, the larger the impact on the infrastructure system and thereby on the consumers due to the disruptive event. The EOH metric can effectively capture the response and resilience of the infrastructure system (Eq. (11)), according to the serviceability criteria chosen by the user.

Similar to EOH of a system, the consumer-level EOH can also be quantified, which indicates the equivalent duration of infrastructure service outage experienced by each consumer (Eq. (12)).

$$\gamma_i^K = \int_{T_0}^T \left[1 - \frac{s_i^K(t)}{S_i^K(t)} \right] dt \quad (12)$$

Finally, in order to compute the resilience of the interdependent infrastructure network, a weighted EOH metric is derived (Eq. (13)).

$$\bar{\gamma} = \sum_{K \in \mathbb{K}} w^K \gamma^K \quad (13)$$

By default, equal weights are applied to both water and power systems.

4. Experiment testbeds

The InfraRisk package is equipped with two self-standing experiment testbeds to carry out generic infrastructure resilience studies, namely, the simple network and the modified Micropolis network. Both networks consist of water-, power-, and traffic networks, and their functional interdependencies.

4.1. Simple network

As the name indicates, the simple integrated network is developed by combining relatively smaller power-, water-, and traffic systems. The networks are constructed using the most basic infrastructure components. The power system consists of three loads, one external grid connection, and five powerlines. The water system consists of 12 pipelines, nine demand nodes, one water pump, and a tank. The traffic system consists of 22 road links, and nine trip generator- and attractor zones. The details related to the infrastructure components in the integrated network are presented in Table 3.

4.2. Modified micropolis network

The second integrated infrastructure test-bed is based on the well-known Micropolis virtual city developed at the Texas A&M University. It is a hypothetical city of 5000 inhabitants and is equipped with residential areas, industrial units, and a central business district. Micropolis was originally developed for the design and simulation of water distribution systems (Brumbelow, Torres, Guikema, Bristow, & Kanta, 2007) due to the unavailability of real network data. Later its scope was extended to other infrastructure systems, including power system and communications (Bagchi, Sprintson, & Singh, 2009; Ciornei et al., 2017) to investigate the role of interdependencies among infrastructure systems. Currently, the testbed consists of water-, power, and transpor system (Fig. 4). The component-level details of the network are presented in Table 4.

5. Experiment simulation

In this section, an experimental simulation to demonstrate the scope of the InfraRisk package is presented. The simulation is performed in the modified Micropolis network (Fig. 4). The objective of the experimental simulation is to quantify the direct and indirect component-level operational risks to infrastructure systems resulting from hypothetical floods along a water stream in the Micropolis city. The direct risks are evaluated using the hazard initiation module, whereas the interdependent simulations are performed to derive the indirect risks. Statistical analysis is then performed to determine the effectiveness of various heuristics-based resilience strategies.

5.1. Hypothetical floods and direct impacts on infrastructure components

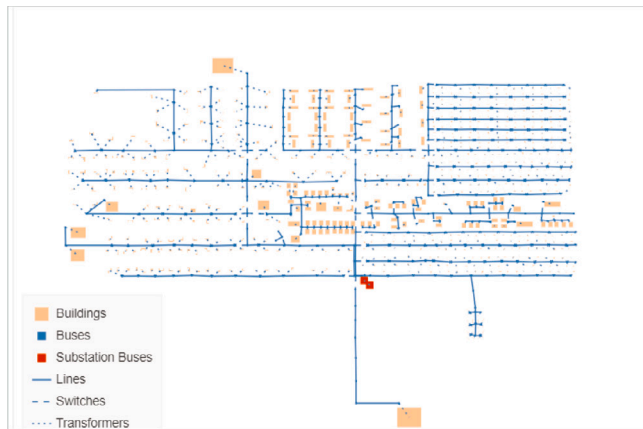
To generate hypothetical floods, a flood hazard profile is required. However, since Micropolis is a virtual city, there is no real historical disaster data to derive the flood risks for the network. Therefore, the authors generated the floods with assumptions on a few of the disaster parameters. It is assumed that the floodplains (regions exposed to floods) are within 100 m from the centerline of the main water stream. Three flood intensities are considered, namely, low, moderate, and high with respective probabilities of occurrence of 0.1, 0.3, and 0.5. The conditional probability of exposure ($p(\text{exposure}|\text{hazard})$) is determined based on the distance criterion as explained in Section 3.2.2. The conditional component failure probabilities ($p(\text{failure}|\text{exposure})$) corresponding to low, moderate, high, and extreme intensity floods generated in the study are 0.1, 0.3, and 0.6, respectively. A total of 165 flood events are simulated, and the corresponding network component failures are determined using Eq. (4). Only water main pipelines (52 nos.), power lines (22 nos.), and road links (17 nos.) are considered for failure as these are the most critical components of the respective infrastructure systems and are exposed to the simulated floods. For the water pipelines, a failed state represents a leakage with area equivalent to half of the cross-sectional area of respective pipe. In the case of power lines and road links, a full failure is assumed. Fig. 5 shows the failure probabilities of exposed infrastructure components to the simulated floods.

5.2. Simulation of interdependent effects

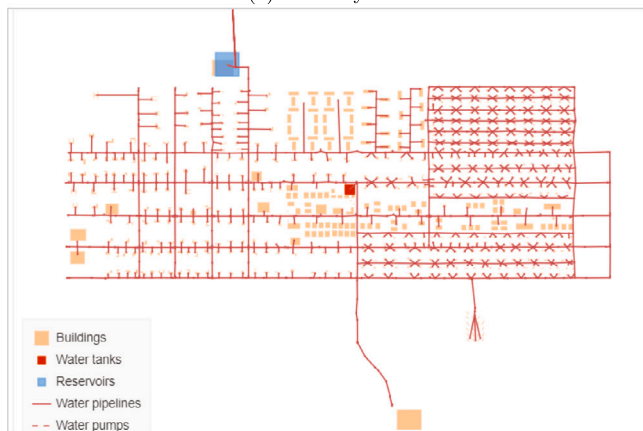
The next step is to perform the simulation of network-wide impacts of the initial infrastructure disruptions due to each flood event. The network-wide impacts are simulated under the three recovery strategies (based on maximum resource-flow handled, betweenness centrality, and land use/zoning). The repair sequences based on each strategy are derived by the recovery module and the resultant repair actions are scheduled and implemented in separate simulations. The prioritized component serviceability (PCS) is used as the measure of performance of systems.

Table 4
Component details of modified Micropolis network.

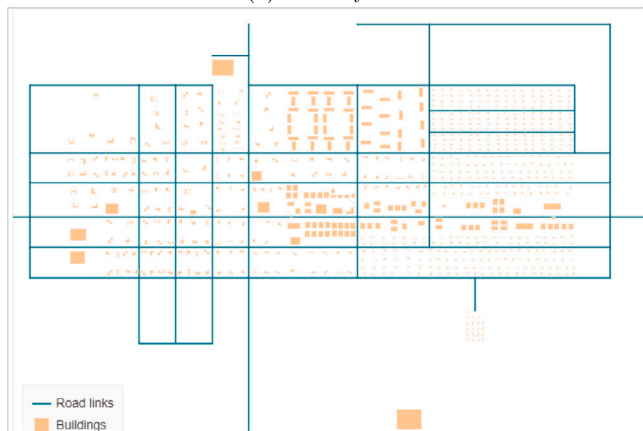
Power system		Water system		Traffic system	
Component	Count	Component	Count	Component	Count
Buses/poles	1225	Main pipelines	678	Arterials	24
Loads	788	Building connections	685	Subarterials	25
Motors	3	Hydrants	52	Collector streets	56
Switches	61	Valves (modeled as pipes)	196	Attractor zones	39
External grids	1	Pumps	8	Generator zones	29
Generators	0	Tanks	1		
Power lines	378	Reservoirs	2		
Transformers	789				



(a) Power system



(b) Water system



(c) Road system

Fig. 4. Topology details of the Micropolis integrated power-water-transport testbed.

Table 5
Repeated measures ANOVA results.

Infrastructure	Source	SS	df	MS	F	p-value
Water	Strategy	2161.12	2	1080.56	12.82	<0.001
	Error	23 927.90	284	84.25		
Power	Strategy	8.17	2	4.08	2.64	0.07
	Error	439.08	284	1.54		

The consumer-level EOH values are evaluated under various strategies for all the simulated flood scenarios. Fig. 6 illustrates the mean EOH of the water and power consumers resulting from the simulated floods when centrality-based restoration strategy was adopted.

5.3. Analysis of the effectiveness of recovery strategies

To statistically test if the three recovery strategies vary in terms of their effectiveness in bringing back the integrated network to the pre-disaster state, repeated measures Analysis of Variance (ANOVA) combined with post-hoc pairwise t-tests are employed. Repeated measures ANOVA is a technique used to detect any statistical difference in the means of related (not independent) groups. The results of the repeated measures ANOVA for both water and power systems are presented in Table 5.

In the case of the water system, the repeated measures ANOVA rejected the null hypothesis in favor of the alternate hypothesis that the expected EOH significantly differed across the recovery strategies [$F(2, 284) = 12.82, p < 0.05$]. The results indicate that there is a significant difference in the effectiveness of recovery strategies in minimizing the cumulative disaster impacts on the consumers. However, in the case of power system, the repeated measures ANOVA did not reject the null hypothesis that the expected EOH across strategies are not significantly different. [$F(2, 284) = 2.64, p > 0.05$].

Motivated by the results of the repeated measures ANOVA tests, post-hoc pairwise t-tests using Benjamini and Hochberg's correction are employed to identify the most-effective recovery strategy for the infrastructure systems (Table 6). In the case of the Micropolis water system, the results show that the expected EOH is the highest for centrality-based strategy (mean = 37.88 h) and the lowest for capacity-based strategy (mean = 31.80 h). The expected EOH of zone-based strategy is 35.71 h. The results suggests the higher effectiveness of capacity-based strategy for Micropolis water system compared to that of centrality- and zone-based strategies. On the other hand, for the power system, the post-hoc tests confirm the findings from the repeated measures ANOVA that there is no statistical difference in the expected EOH among strategies based on capacity (mean = 2.98 h), centrality (mean = 2.90 h), and zone (mean = 3.23 h). The higher mean EOH values of water system compared to that of power system confirms that the latter is more resilient against the flood impacts.

The results indicate that each infrastructure system may have specific repair strategies that are the most effective, even when the amounts of resources available for recovery are similar. This could be attributed to the unique topological, functional, and recovery characteristics of the infrastructure systems.

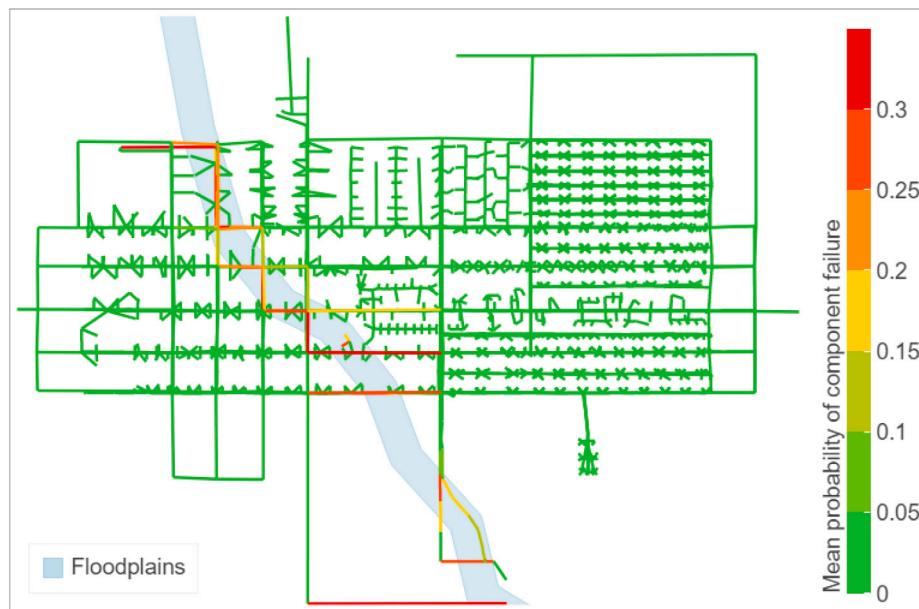


Fig. 5. Probability of failure of infrastructure components (all three infrastructure systems) based on the simulated floods.

Table 6

Post-hoc tests for pairwise comparison of effectiveness of recovery strategies.

Infrastructure	Strategy 1	Strategy 2	Mean diff.	T	p-value
Water	Capacity	Centrality	−6.08	−5.53	<0.0001
	Capacity	Zone	−3.91	−3.69	<0.0001
	Centrality	Zone	2.16	2.21	0.002
Power	Capacity	Centrality	0.09	0.89	0.372
	Capacity	Zone	−0.24	−1.35	0.269
	Centrality	Zone	−0.33	−2.11	0.109

5.4. Model limitations

While InfraRisk offers a wide-range of applications in infrastructure resilience research, it also has a few shortcomings and limitations, which could be addressed in future research. Some of these limitations are listed below.

- Since InfraRisk has integrated existing infrastructure simulators, the limitations of each simulator must also be considered while inferring the network simulation outcomes.
- The hazard module in InfraRisk currently models exposure, vulnerability, and consequence (failure of components) purely from a probability perspective. These aspects could be better modeled by incorporating disaster-dependent and infrastructure-specific fragility curves, leading to more reliable simulation results.
- There could be additional interdependencies, which may be relevant to risk and resilience assessment of infrastructure systems, depending upon the scope of the study.
- The scope of the current model is limited to the power-, water- and road- system. In future research, additional interdependent critical infrastructure could be modeled (e.g., communication network).

6. Conclusions

This research presented an open-source simulation package for simulating disaster impacts on interconnected power-, water-, and traffic systems and analyze their network-wide effects. Existing infrastructure-specific simulation models are integrated using a sequential simulation framework for interdependent infrastructure simulations. The details of the constituent modules developed for the integrated infrastructure

network modeling, hazard initiation and vulnerability modeling, recovery modeling, simulation of direct and indirect disaster impacts, and resilience quantification, are discussed. Later, the features and capabilities of the simulation package are demonstrated using a case study based on a virtual integrated network. While the infrastructure model can simulate the component-level performance of the infrastructure systems considering the interdependencies, the resilience indices are designed to track the quality of infrastructure services at the consumer-level. The above features make the simulation platform unique and capable of testing realistic resilience policies, strategies, and interventions in a controlled environment. Some of the potential applications of the package are listed below.

- Assessment of pre-disaster component- and system-level resilience enhancement measures, such as structural enhancements.
- Identification of system-level vulnerabilities and critical components.
- Post-disaster recovery enhancement by improving existing restoration strategies and protocols, resource allocation, etc.
- Resilience-based design of greenfield urban infrastructure systems.

Beyond interfacing of existing infrastructure simulators, InfraRisk offers a single platform for hazard generation, interdependency modeling, recovery modeling, and resilience quantification, allowing detailed resilience analysis of networks and constituent infrastructure components. The object-oriented nature would also help users to use more accurate vulnerability and recovery models along with InfraRisk to gather more accurate insights into resilience of interdependent power-, water-, transport networks

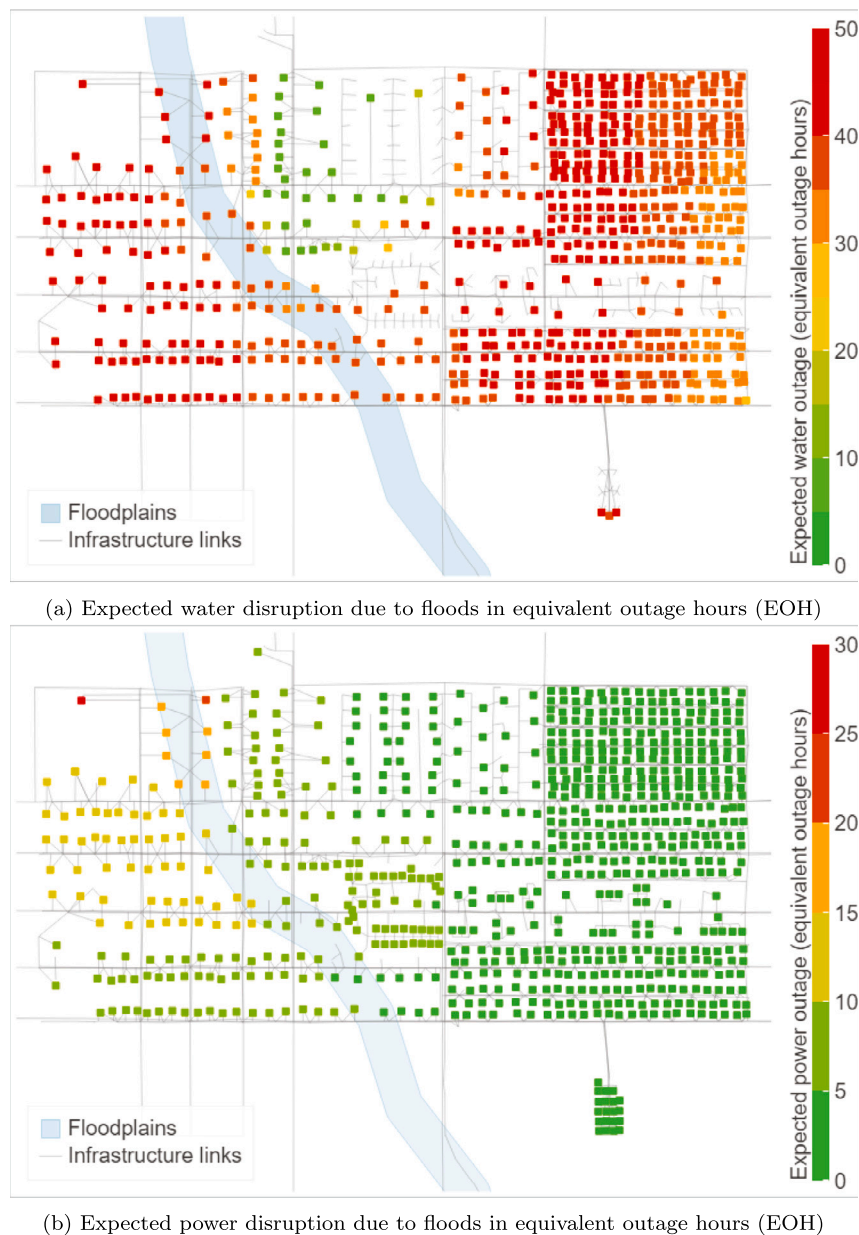


Fig. 6. Expected consumer-level impacts for centrality-based repair strategy.

CRedit authorship contribution statement

Srijith Balakrishnan: Conceptualization, Methodology, Data curation, Software, Visualization, Formal analysis, Writing – original draft, Project administration. **Beatrice Cassottana:** Conceptualization, Methodology, Validation, Resources, Writing – review and editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The InfraRisk package can be downloaded from [Github](https://github.com). Documentation and codes for sample simulations are available in the InfraRisk package.

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